Uniform and Log-normal Distributions

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Probability distributions.
- **Leads to:** Function transformer; Beam model; Uniform random rotations and directions; Probabilistic roadmap (PRM).
- **Compare with:** Pareto distribution and power laws.

1. Overview

Key point: Beyond the normal distribution, three continuous distributions appear constantly: the flat uniform, the right-skewed log-normal, and the power-law Pareto; this Note covers the first two.

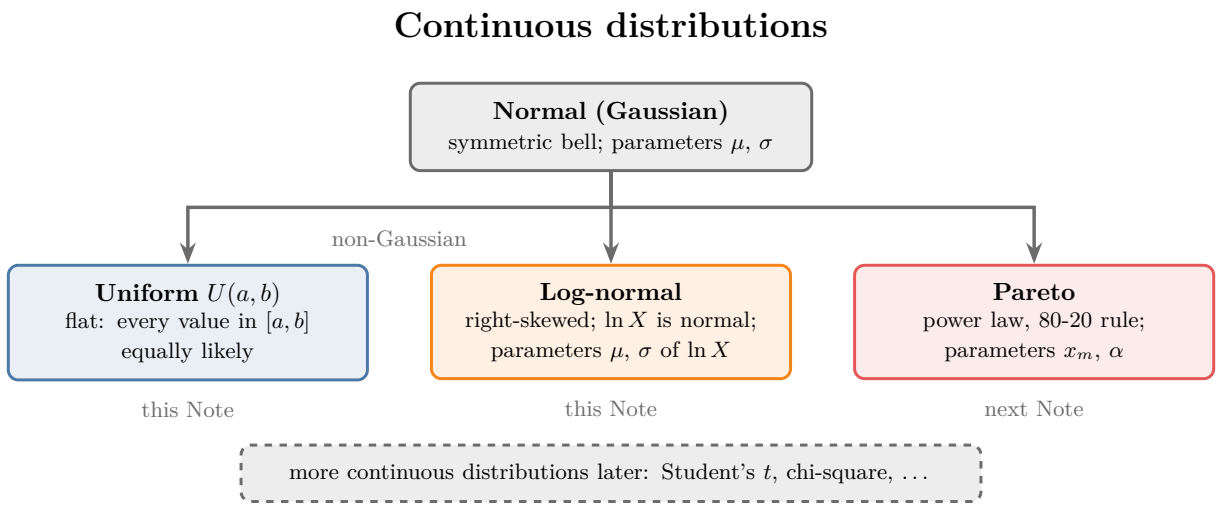


Figure 1: The continuous distributions of this Note and the next, beside the normal distribution

The normal distribution (see what the normal distribution is) is the best studied distribution, but there are hundreds of others. Distributions that are not normal are called **non-Gaussian** (G-1334). Figure 1 shows the three famous continuous ones of this Note and the next (they were first listed in famous distributions).

For each one we look at:

- its definition;
- its PDF and parameters;
- where it shows up;

- how to recognise it in data. Recognising uses the Q-Q plot (a plot that compares the sorted data with the values a chosen distribution would give).

2. The uniform distribution

> **Key point:** In a uniform distribution every outcome in a given range is equally likely; it comes in a discrete and a continuous version.

The **uniform distribution** (G-2043) is a probability distribution in which all outcomes are equally likely within a given range. If we pick a random value from that range, any value is as likely as any other.

The uniform distribution comes in two kinds, one for each kind of random variable:

- **Discrete uniform** (G-618): a fair die. Each face from 1 to 6 has probability 1/6, so the PMF (the probability of each value) is six bars of equal height (see the PMF of one die).
- **Continuous uniform** (G-467): a continuous random variable spread evenly over a range, such as a production time anywhere between 5 and 6 hours. This Note is about this kind.

2.1 Notation and parameters

> **Key point:** $X \sim U(a, b)$: the two parameters are the lower end a and the upper end b of the range.

The idea in plain words: a machine takes anywhere between 5 and 6 hours to make one product, and every time in that range is equally likely. Only two numbers describe it: the shortest time, 5, and the longest time, 6.

Those two numbers are the **parameters** of the **continuous uniform distribution** (G-467). The short way to write the machine's time X is:

$$X \sim U(5, 6)$$

The symbol $\sim$ reads "follows the distribution", and $U(5, 6)$ is the **uniform distribution** (G-2043) from 5 to 6. In general we write:

$$X \sim U(a, b)$$

Here a is the lowest possible value (5 for the machine) and b the highest (6 for the machine), with $b > a$.

2.2 The PDF of the continuous uniform

> **Key point:** The PDF is a flat line at height $1/(b-a)$ between a and b, and 0 everywhere else.

The graph of the PDF is a rectangle (Figure 2, left). The rule that fixes its height: the total area under every PDF (probability density function) is 1 (see area under the curve is probability).

For the machine, $X \sim U(5, 6)$:

$$\text{width} = 6 - 5 = 1 \text{ hour}$$

$$\text{area} = \text{width} \times \text{height} = 1$$

$$\text{height} = \frac{1}{1} = 1 \text{ per hour}$$

The probability that a product takes between 5.2 and 5.5 hours is the area of the part of the rectangle between those two times:

$$\text{width} = 5.5 - 5.2 = 0.3$$

$$P(5.2 \leq X \leq 5.5) = 0.3 \times 1 = 0.3$$

Figure 2 (left) shades exactly this rectangle; Figure 2 (right) shows the CDF (the share of products finished by each time). The same steps for any range give the **PDF** of the **continuous uniform** (G-467). The rectangle is $b - a$ wide, so its height must be $1/(b - a)$:

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

For $a = 5$ and $b = 6$ the height is 1, as in the steps above:

$$1/(6 - 5) = 1$$

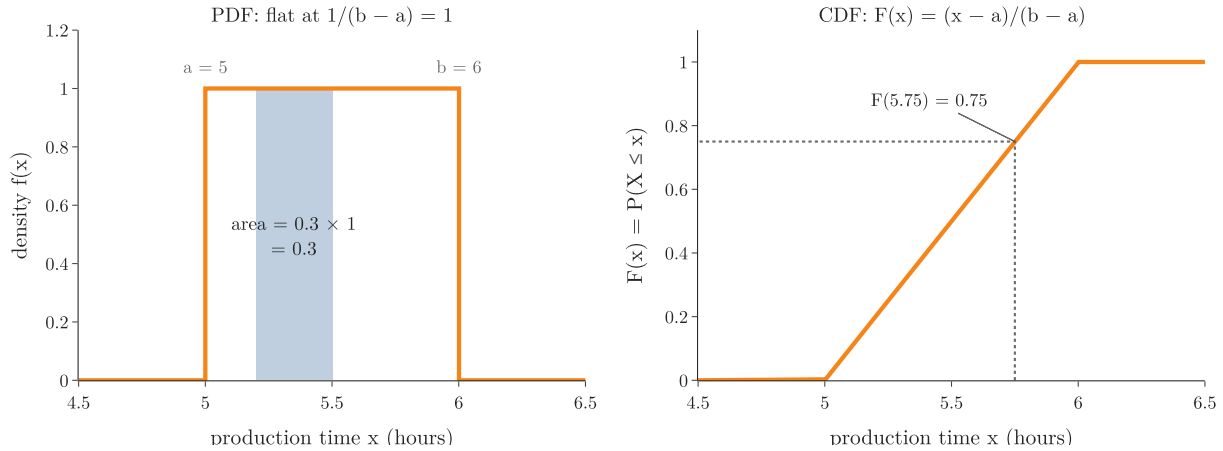


Figure 2: U(5, 6): the flat PDF with P(5.2 ≤ X ≤ 5.5) shaded (left), and the CDF (right)

The continuous uniform distribution is **symmetric**, like the normal distribution: its skewness is 0. Its excess kurtosis is −1.2, as it has no tails at all (see **excess kurtosis** (G-720)).

Extra: The CDF, mean and variance of $U(a, b)$.
The idea: area builds up at a constant rate, so the share finished is a straight ramp; the mean is the midpoint; the variance grows with the square of the width. For $U(5, 6)$, in steps:

$$F(5.75) = \frac{5.75 - 5}{6 - 5}$$

$$F(5.75) = 0.75$$

(the dotted lines of Figure 2, right).

$$\text{mean} = \frac{5 + 6}{2} = 5.5 \text{ hours}$$

$$\text{variance} = \frac{(6 - 5)^2}{12} = 0.0833$$

$$\text{standard deviation} = \sqrt{0.0833} = 0.289 \text{ hours}$$

The general formulas behind those lines, for $a \le x \le b$:

$$F(x) = \frac{x - a}{b - a}$$

$$\text{mean} = \frac{a + b}{2}$$

$$\text{variance} = \frac{(b - a)^2}{12}$$

The CDF is a straight ramp from 0 at a to 1 at b.

2.3 Where the continuous uniform appears

Key point: The **continuous uniform** (G-467) appears when a quantity is equally likely anywhere in a fixed range, and it works behind the scenes in many ML techniques.

Everyday examples are quantities that are equally likely anywhere within fixed limits:

- the time a machine takes to produce a product, when it ranges evenly from 5 to 6 hours;
- the waiting time for a bus that comes exactly every 10 minutes, for someone arriving at a random moment: anywhere from 0 to 10 minutes.

In machine learning, the **uniform distribution** (G-2043) mostly works behind the scenes:

- **Random initialization** (G-1612). Neural networks and k-means clustering (see **k-means** (G-996)) start from random parameter values and improve them step by step. The start strongly affects the final result. Drawing the start from a uniform distribution gives every value in the range the same chance.
- **Sampling.** Splitting data into training and test sets, or drawing a random subset, picks each **observation** (G-1374; one record, one row of the data table) with equal probability (the train-test split).
- **Data augmentation** (G-531). In deep learning with images, a small dataset is enlarged by making new images from old ones: zoomed in a little, shrunk, shifted, rotated. The amounts are drawn at random, often uniformly.
- **Hyperparameter tuning** (G-909). Random search tries hyperparameter values drawn from ranges, often uniformly (scikit-learn `RandomizedSearchCV` docs; see randomized search).
- **Pseudo-random number generators.** Computers first produce uniform random numbers and turn them into samples from other distributions (MML §6.7.1).

Python: Uniform values and the uniform distribution.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
times = rng.uniform(5, 6, size=1000)   # from 5 to 6
# scipy uses loc = a and scale = b - a
u = stats.uniform(loc=5, scale=1)
u.pdf(5.3)                  # 1.0
u.cdf(5.5) - u.cdf(5.2)     # 0.3
```

A common slip: `stats.uniform(5, 6)` means $U(5, 11)$, because the second number is the width, not the upper end.

3. The log-normal distribution

> **Key point:** A random variable is log-normal when its logarithm is normally distributed; the variable itself is right-skewed with a long tail.

The idea in plain words: some quantities grow by multiplying, not by adding. A forum comment is not "20 words more" than a short one; it is "8 times longer". Such quantities are bunched at small values with a long tail of big ones. The log measures a value by how many times it was multiplied, so "8 times longer" and "8 times shorter" become steps of the same size (see the log transform).

Take the comment lengths of section 3.4, with a median of about 20 words. Compare a comment 8 times longer and one 8 times shorter:

	words	distance from 20 words	ln of words	distance from $\ln 20 = 2.996$
8 times shorter	2.5	17.5	0.916	2.080 below
median	20	0	2.996	0
8 times longer	160	140	5.075	2.079 above

In words the two comments sit very unequal distances from the median (17.5 and 140); after the log they sit the same distance (about 2.08) on either side. That is why the long right tail becomes a symmetric bell.

Quantities like this follow a **log-normal distribution** (G-1115): a heavy-tailed continuous distribution of a random variable whose logarithm is normally distributed. Two things define it:

1. **The data is right-skewed:** many small values and a long tail of large ones.
2. **The log of the data is normal:** take the natural log (ln, the logarithm to base $e \approx 2.718$) of every value and plot the new values; they form a bell curve.

The second condition is the test. Not every right-skewed distribution is log-normal, only one whose logs come out normal. In symbols, with $\iff$ reading "if and only if":

$$X \text{ is log-normal} \iff \ln X \text{ is normal}$$

Figure 3 applies the log gradually to 1,000 simulated comment lengths (the data of section 3.4). It uses the power $(x^\lambda - 1)/\lambda$, which leaves the shape of the raw data unchanged at $\lambda = 1$ and becomes $\ln x$ as λ shrinks to 0. Watch the long right tail pull in: the skewness falls from 4.41 to 1.09, then 0.59, and the logs form a bell with skewness -0.02.

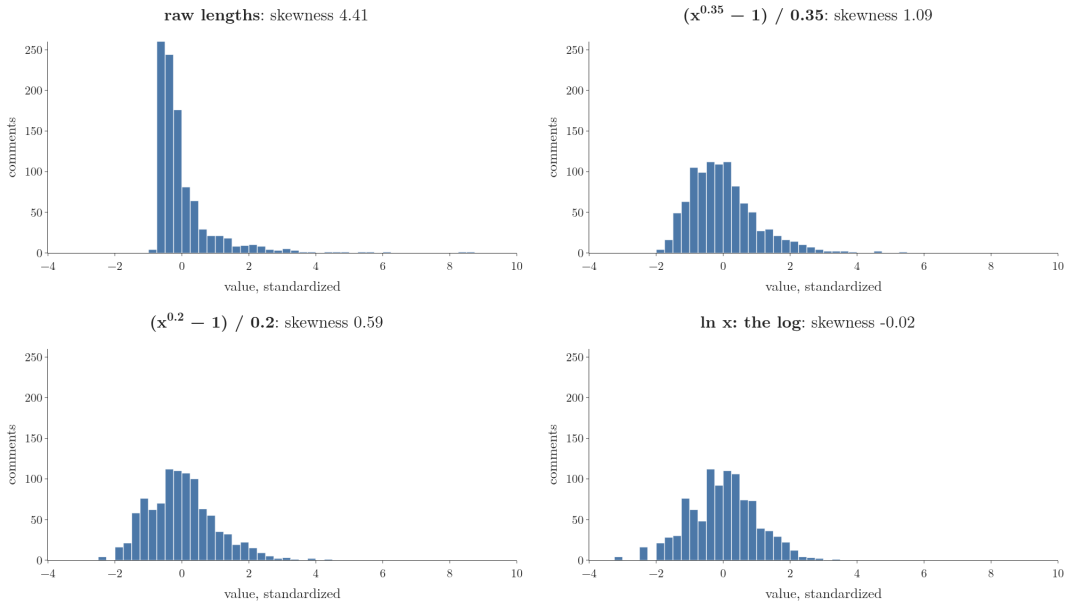


Figure 3: 1,000 simulated comment lengths transformed step by step from raw ($\lambda = 1$) to the log ($\lambda = 0$), each standardized to one axis. The right-skewed histogram becomes a bell.

3.1 Notation and parameters

> **Key point:** The two parameters μ and σ are the mean and standard deviation of $\ln X$, not of X.

For the comment lengths, the logged values are bell-shaped around 3 with a spread of 1. So the numbers that describe the logged values are $\mu = 3$ (their mean) and $\sigma = 1$ (their standard deviation); the word counts themselves have a median of $e^3 = 20.1$ words.

We write this as

$$X \sim \text{Lognormal}(\mu, \sigma^2)$$

which means

$$\ln X \sim N(\mu, \sigma^2)$$

where $N(\mu, \sigma^2)$ is the normal distribution with mean μ and variance σ^2 (for the comments, $N(3, 1)$). The parameters μ and σ look like those of the normal distribution, and they are: but they belong to the logged values. The mean and standard deviation of X itself are different numbers.

Figure 4 keeps $\mu = 0$ and increases σ. A larger σ spreads the curve out and stretches the right tail. All three curves start at 0: a log-normal variable is always positive.

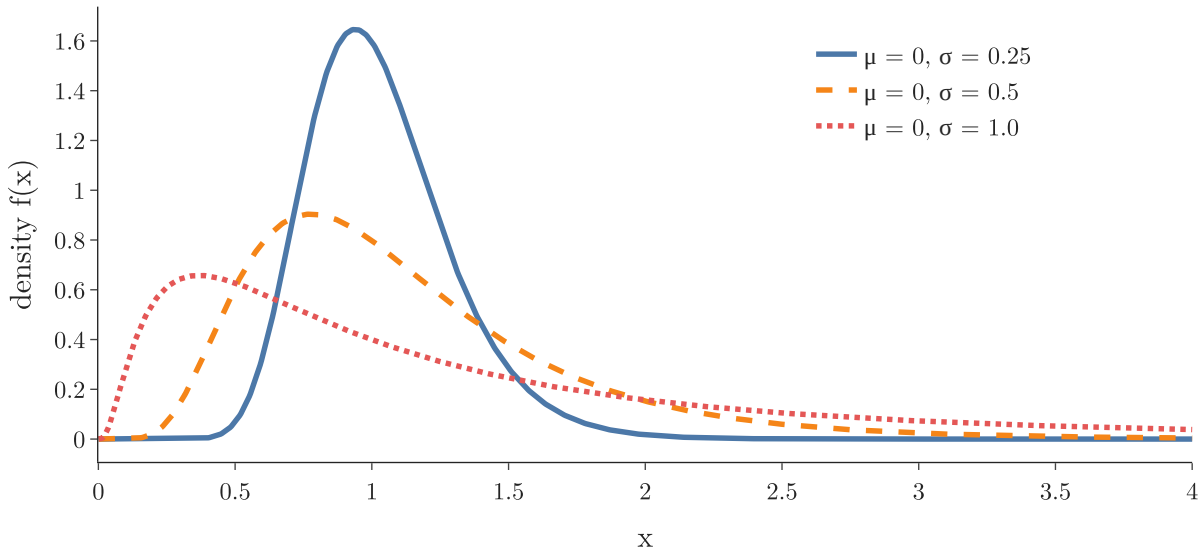


Figure 4: Log-normal PDFs with $\mu = 0$ and three values of σ

3.2 The log-normal PDF

> **Key point:** The log-normal PDF is the normal PDF with $\ln x$ in place of x, divided by an extra x.

The idea: take the normal PDF, put $\ln x$ where x was, and divide by x. The extra x comes from the log transformation: the log squeezes large values together, and the factor $1/x$ keeps the total area at 1 (the change-of-variables rule, MML §6.7.2).

Worked case: comment lengths on a forum with $\mu = 3$ and $\sigma = 1$ (in log-words), at $x = 20$ words:

$$\ln 20 = 2.996$$

$$\ln 20 - \mu = 2.996 - 3 = -0.004 \approx 0$$

$$\text{exponent} = -\frac{(-0.004)^2}{2 \times 1^2} \approx 0$$

$$e^0 = 1$$

$$x\,\sigma\sqrt{2\pi} = 20 \times 1 \times 2.5066 = 50.13$$

$$f(20) = \frac{1}{50.13} \times 1 = 0.0199$$

The same recipe for any $x > 0$ is the **log-normal PDF**:

$$f(x) = \frac{1}{x\,\sigma\sqrt{2\pi}}\, e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

Each line of the worked case is one piece of this formula: the exponent, the factor $x\sigma\sqrt{2\pi}$, and their combination.

The curve looks like a normal curve pushed to the left with its right side stretched out (Figure 4). The CDF behaves like the normal CDF too: a larger σ makes it rise more slowly.

The resemblance is only in the formulas. The log-normal variable itself is skewed and not symmetric; it is $\ln X$ that is normal and gets all the benefits of normality.

Extra: Because $\ln X$ is normal, every log-normal probability is a normal probability in disguise.
The idea: take the log of the cut-off, standardize it with μ and σ, and read the normal CDF Φ ($\Phi(z)$ is the share of a standard normal at or below z, for example $\Phi(0) = 0.5$). For the comments, the share longer than 100 words:

$$\ln 100 = 4.605$$

$$z = \frac{4.605 - 3}{1} = 1.605$$

$$\Phi(1.605) = 0.946$$

$$P(X > 100) = 1 - 0.946 = 0.054$$

The median comment has $e^3 = 20.1$ words, and the mean is $e^{3.5} = 33.1$: the long right tail pulls the mean above the median. The general formulas behind these lines:

$$P(X \le x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$$

$$\text{median} = e^{\mu}$$

$$\text{mean} = e^{\mu + \sigma^2/2}$$

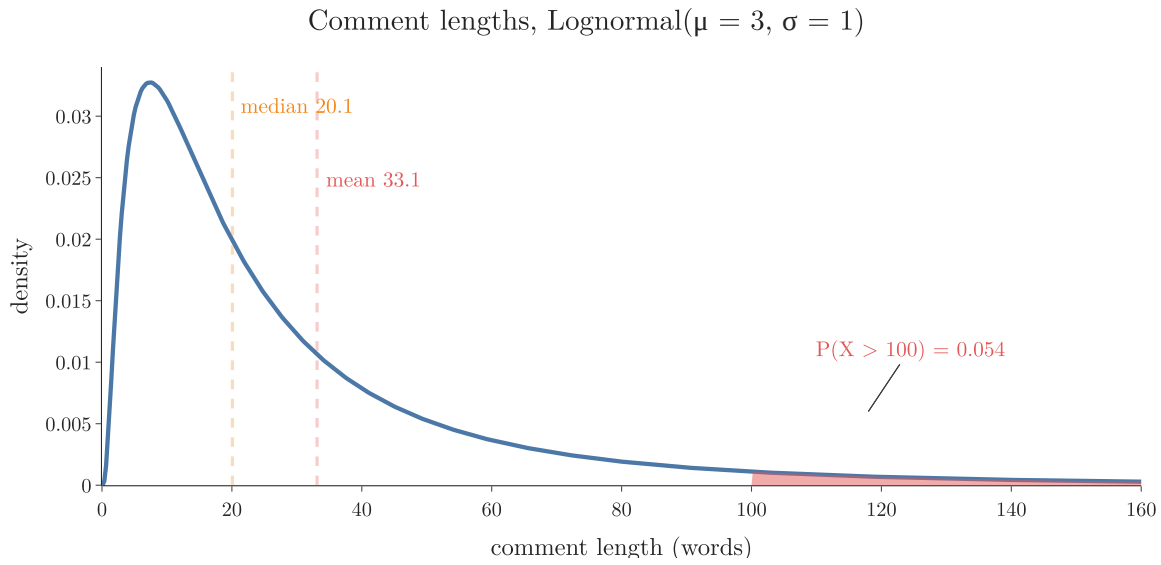


Figure 5: The comment-length log-normal ($\mu = 3$, $\sigma = 1$): median 20.1, mean 33.1, and the 5.4 percent of comments longer than 100 words shaded.

3.3 Where the log-normal appears

> **Key point:** Log-normal data is common in internet applications: comment lengths, time spent on articles, and many human behaviours.

The **log-normal distribution** (G-1115) appears in biology, medicine, chemistry, hydrology and economics, and it is especially common in internet applications:

- **Length of comments** in online discussion forums: most comments are a few words; a few are very long.
- **Time spent reading online articles:** many readers leave quickly; a few stay to the end and read the other comments too.
- **Length of chess games:** many games end quickly, and a few go on for a very long time.
- **Income:** there is evidence that the income of about 97 to 99% of the population is log-normally distributed: many people earn little, few earn a lot. The richest 1 to 3% follow a **Pareto distribution** (G-1453) (Clementi and Gallegati 2005).

3.4 How to check whether data is log-normal

> **Key point:** Take the log of every value and check the logs for normality with a Q-Q plot; if the logs are normal, the data is log-normal.

Checking for log-normality is a common interview question, and the answer follows from the definition:

1. Take the natural log of every value of X; call the result $Y = \ln X$.

2. Draw a Q-Q plot of Y against the normal distribution (see building a Q-Q plot).
3. If the points lie on a straight line, Y is normal, so X is log-normal.

Figure 6 does this for 1,000 comment lengths simulated with $\mu = 3$ and $\sigma = 1$. The raw lengths are strongly right-skewed (skewness 4.4). Their logs form a bell (skewness -0.02), and the Q-Q plot of the logs is a straight line.

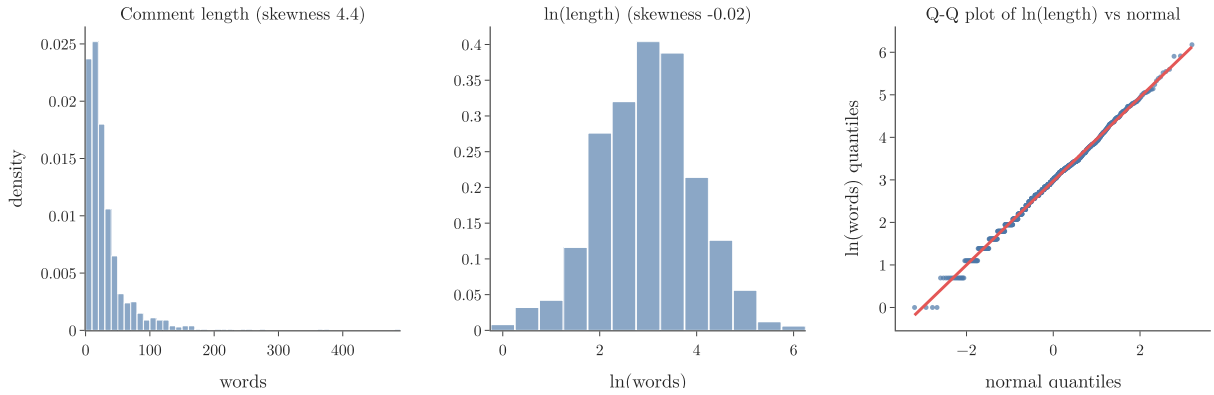


Figure 6: 1,000 comment lengths: raw (left), after the log (middle), and the Q-Q plot of the logs (right)

The same check shows the payoff of knowing a **feature** (G-772; one variable of the data, one column of the table) is log-normal: the log transform turns it into a normal feature, and everything that works on normal data then applies. That transform, and how it helps models such as linear and logistic regression, is in the function transformer.

Python: Log-normal values, and the scipy parameters.

```
# mean and sigma here are mu and sigma of ln X
words = rng.lognormal(mean=3, sigma=1, size=1000)
logs = np.log(words)
stats.probplot(logs, dist="norm")      # Q-Q plot data
# scipy: s = sigma, scale = e^mu
ln = stats.lognorm(s=1, scale=np.exp(3))
ln.pdf(20)        # 0.0199
ln.sf(100)        # 0.054, P(X > 100)
```

The Notebook (`MA-029-uniform-and-log-normal.ipynb`) rounds the simulated lengths to whole words and draws Figure 6 with Plotly.

4. Summary

	Uniform $U(a, b)$	Log-normal
Shape	flat between a and b	right-skewed, long right tail
Parameters	a, b (ends of the range)	μ, σ of $\ln X$
PDF	$1/(b-a)$ on $[a, b]$ (Section 2.2)	normal PDF of $\ln x$, divided by x (Section 3.2)
Skewness	0 (symmetric)	positive
Example	production time 5 to 6 hours	comment length, income
Check	Q-Q plot against uniform	Q-Q plot of $\ln X$ against normal
scipy	`uniform(loc=a, scale=b-a)`	`lognorm(s=`σ`, scale=exp(`μ`))`

- Read the table's scipy row with care, because scipy takes the width $b - a$ (not b) for the uniform and e^{μ} for the log-normal, so `uniform(5, 6)` means $U(5, 11)$.
- **Non-Gaussian** (G-1334) simply means not normal, so the normal distribution's rules do not carry over to it.
- The **uniform distribution** (G-2043) works behind the scenes in ML: initialization, sampling, augmentation, random search, because each needs every value in a range, or every row, to have the same chance.
- A right-skewed feature is log-normal only if its logs are normal, so check with a Q-Q plot of the logs, not the skew alone.
- Log-normal data becomes normal with a log transform, so everything that works on normal data then applies to it.
- This answers the opening question for two of the three common non-normal distributions: the uniform is flat, the log-normal is right-skewed with normal logs.

5. Sources

Built from

- CampusX, "Session 42 - Non-Gaussian Probability Distributions | DSMP 2023", YouTube, https://www.youtube.com/watch?v=U6QCc_3zgUk

Other references

- Clementi, F. and Gallegati, M. (2005). "Pareto's Law of Income Distribution: Evidence for Germany, the United Kingdom, and the United States." In *Econophysics of Wealth Distributions*, Springer.
- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Sections 6.7.1 (sampling with the inverse CDF) and 6.7.2 (change of variables).
- scikit-learn documentation, `RandomizedSearchCV`.

6. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Non-Gaussian distribution (G-1334)	Any distribution that is not normal.
Uniform distribution (G-2043)	A distribution in which every outcome in a range is equally likely.
Continuous uniform distribution $U(a, b)$ (G-467)	A continuous variable spread evenly between a and b, with density $1/(b-a)$.
Random initialization (G-1612)	Starting a model's parameters at random values, often drawn from a uniform distribution.
Log-normal distribution (G-1115)	A distribution with a long tail to the right (right-skewed, continuous) whose values follow a normal distribution once you take their logarithm.
Lognormal(μ, σ^2) (G-20)	A short way to write a log-normal distribution: positive values with a long tail to the right (right-skewed), whose logarithm is normal, $\ln X \sim N(\mu, \sigma^2)$.
Discrete uniform distribution (G-618)	A discrete distribution in which every possible value is equally likely, such as a fair die.
Excess kurtosis (G-720)	Kurtosis minus 3, so a normal distribution scores 0.
k-means (G-996)	A clustering algorithm that repeatedly assigns points to the nearest centre and moves each centre to the mean of its points.
Observation (G-1374)	One record of a dataset: one row of the data table.
Data augmentation (G-531)	Enlarging a dataset by making changed copies of its examples, such as zoomed or shifted images.
Hyperparameter tuning (G-909)	Trying several hyperparameter values and keeping the best.
Pareto distribution (G-1453)	A distribution of values that start at a smallest value x_m and become rarer the larger they are, in a long right tail (a right-skewed power law); used to model wealth and income. PDF $\alpha x_m^{\alpha}/x^{\alpha+1}$.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).